Clinical Infectious Diseases

IDSA GUIDELINES

2025 Clinical Practice Guideline Update by the Infectious Diseases Society of America on the Treatment and Management of COVID-19: Infliximab

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This article provides a focused update to the clinical practice guideline on the treatment and management of patients with coronavirus disease 2019 (COVID-19), developed by the Infectious Diseases Society of America. The guideline panel presents a recommendation on the use of infliximab in hospitalized adults with severe or critical COVID-19. The recommendation is based on evidence derived from a systematic literature review and adheres to a standardized methodology for rating the certainty of evidence and strength of recommendation according to the GRADE (Grading of Recommendations, Assessment, Development, and Evaluation) approach.

Keywords. COVID-19; SARS-CoV-2; infliximab; monoclonal antibody; guideline.

In patients hospitalized with severe or critical COVID-19 receiving systemic glucocorticoids, should infliximab compared to no infliximab be added to standard care?

Recommendation: In hospitalized adults receiving systemic glucocorticoids who are experiencing severe, rapidly progressing COVID-19* or critical COVID-19**, when baricitinib and tocilizumab are not available, the IDSA guideline panel suggests infliximab rather than no infliximab (*conditional recommendation, low certainty of evidence*).

*Severe, rapidly progressing illness is defined as patients with SpO_2 ≤94% on room air, including patients on supplemental oxygen, who are worsening despite treatment with systemic glucocorticoids.

**Critical illness is defined as patients requiring high-flow nasal cannula oxygen/noninvasive ventilation or invasive mechanical ventilation or ECMO.

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The guidelines were posted online at https://www.idsociety.org/practice-guideline/COVID-19-guideline-treatment-and-management/ on 30 May 2025. As COVID-19 treatment and management guidelines may change rapidly with evolving virus variants and ongoing research, please check the website for the most current version of this guideline.

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BACKGROUND

Infliximab is a chimeric monoclonal antibody that targets tumor necrosis factor alpha (TNF-α), a pro-inflammatory cytokine mainly generated by activated macrophages, lymphocytes, and natural killer cells. By neutralizing TNF-α, infliximab reduces inflammation and immune system overactivation, making it effective for autoimmune and inflammatory conditions [1].

Infliximab is U.S. Food and Drug Administration (FDA) approved. Indications include Crohn's disease, ulcerative colitis, rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, and plaque psoriasis [2, 3]. Early in the COVID-19 pandemic, increased levels of interleukin-6 (IL-6) and TNF-α were identified as independent predictors of disease severity and survival [4]. Moreover, several cohort studies noted that people receiving TNF-α inhibitors were at lower risk for COVID-19–related hospitalizations and severe disease in comparison to people who were receiving non-TNF-α biologics [5, 6]. Although infliximab is not FDA-approved for COVID-19, it has been studied for its potential role in managing severe COVID-19–related inflammation (eg, cytokine storm), most notably in the ACTIV-1 trial [7].

In this focused update to the 2023 guideline [8], a recommendation is provided for infliximab. The primary audience for this recommendation is clinicians treating hospitalized adults with severe or critical COVID-19.

METHODS

The panel's recommendation is based upon a systematic review of available evidence and adheres to a standardized methodology for rating the certainty of evidence and strength of recommendation according to the GRADE (Grading of Recommendations, Assessment, Development, and Evaluation) approach (Supplementary Figure 1) [9]. The recommendation has been endorsed by the Pediatric Infectious Diseases Society, Society for Healthcare Epidemiology, and the Society of Critical Care Medicine.

Strong recommendations are made when the recommended course of action would apply to most people with few exceptions. Conditional recommendations are made when the suggested course of action would apply to the majority of people with many exceptions and shared decision making is important.

A literature search was conducted in August 2024 as part of a systematic review. Key eligibility criteria at both the topic and clinical question levels guided the selection of studies for inclusion. For this clinical question, only hospitalized adults were included. The primary comparison of interest was infliximab versus no infliximab.

A critical appraisal of the evidence according to the GRADE approach, along with an assessment of the benefits and harms of care options, informed the recommendation(s) [9, 10]. Details of the systematic review and guideline development processes are available in the Supplementary Material.

SUMMARY OF EVIDENCE

The search identified 2 randomized controlled trials (RCTs) that reported on adults 18 years or older with severe COVID-19 who were randomized to treatment with infliximab (5 mg/kg) plus standard of care or standard of care alone (Supplementary Table 1) [7, 11]. One open-label RCT (N = 69), called CATALYST, treated patients 16 years or older with a single intravenous dose given over 2 hours on day 1 plus standard of care, which consisted of the following: dexamethasone (100%), corticosteroids (94%), remdesivir (29%), and tocilizumab (6%) [11]. Patients in the comparison arm received standard of care alone: dexamethasone (100%), corticosteroids (85%), remdesivir (62%), and tocilizumab (3%). One blinded and placebo-controlled trial (N = 1049), ACTIV-1, treated patients with a single infusion of infliximab plus standard of care compared to standard of care alone [7]. Standard of care across the treatment and control arms included remdesivir (93%–94%), corticosteroids (89%–93%), tocilizumab (3%), and baricitinib (1%–3%). Though the primary endpoints of O'Halloran et al [7] and Fisher et al [11] were time to recovery from COVID-19 pneumonia and improvement in inflammation, respectively, the trials also reported on outcomes of mortality at 28 days, recovery at 28 days (assessed by the first day a hospitalized participant did not require oxygen or ongoing care, or patient was not hospitalized with or without limitations on activities, ie, World Health Organization categories 6, 7, or 8), length of hospitalization, and serious adverse events (Table 1).

BENEFITS

Among hospitalized patients, infliximab reduces mortality at 28 days compared to no infliximab treatment (risk ratio [RR]: 0.70; 95% confidence interval [CI]: 0.51–0.96]; low certainty of evidence). In addition, patients receiving infliximab trended toward improved recovery as measured by improvement to hospitalization without oxygen requirements or release from the hospital (hazard ratio, 1.12; 95% CI: 0.99–1.28; moderate certainty of evidence). The effect of infliximab treatment on hospitalization is very uncertain (mean difference, 1 day fewer; 95% CI: 13.27 days fewer to 11.27 days more; very low certainty of evidence).

HARMS

Serious adverse events among patients receiving infliximab did not differ from those receiving usual care (RR: 0.97; 95% CI: 0.78–1.19; low certainty of evidence). The low certainty of evidence was due to very serious concerns with imprecision, reflecting both the wide CI, which includes appreciable benefits and possible harms, and few reported events.

OTHER CONSIDERATIONS

The panel agreed that the overall certainty of evidence was low (Table 1), given the sparseness in mortality data, risk of bias concerns in the study addressing length of hospitalization (Supplementary Table 2), and because the 95% CIs cannot exclude the potential for no meaningful difference in mortality, recovery, length of hospitalization, and serious adverse events.

Table 1. GRADE evidence profile: In patients hospitalized with severe or critical COVID-19 receiving systemic glucocorticoids, should infliximab compared to no infliximab be added to standard care?

Certainty Assessment							No. of Patients		Effect			
No. of Studies	Study Design	Risk of Bias	Inconsistency	Indirectness	Imprecision	Other Considerations	Infliximab	No Infliximab	Relative (95% CI)	Absolute (95% CI)	Certainty	Importance
Mortality (follow-up 28 d)												
2 [7, 11]	Randomized trials	Not serious	Not serious	Not serious[a]	Very serious[b,c]	None	56/552 (10.1%)	80/550 (14.5%)	RR 0.70 (0.51–0.96)	44 fewer per 1000 (from 71 fewer to 6 fewer)	⨁⨁◯◯ Low	Critical
Recovery (follow-up 28 d; assessed with: first day a hospitalized participant did not require oxygen or ongoing care or patient was not hospitalized with or without limitations on activities)[d]												
1 [7]	Randomized trials	Not serious	Not serious	Not serious[a]	Serious[c]	None	421/531 (79.3%)	405/530 (76.4%)	HR 1.12 (0.99–1.28)[e]	38 more per 1000 (from 3 fewer to 78 more)	⨁⨁⨁◯ Moderate	Critical
Length of hospitalization												
1 [11]	Randomized trials	Serious[f]	Not serious	Not serious[a]	Very serious[b,g]	None	35	34	---	MD 1 d fewer (13.27 fewer to 11.27 more)	⨁◯◯◯ Very low	Critical
SAEs (assessed with: death, life-threatening AE, new/prolonged hospitalization, persistent/significant incapacity/substantial disruption of normal life functions, congenital anomaly/birth defect)												
2 [7, 11]	Randomized trials	Not serious	Not serious	Not serious[a]	Very serious[g]	None	131/552 (23.7%)	135/550 (24.5%)	RR 0.97 (0.78–1.19)	7 fewer per 1000 (from 54 fewer to 47 more)	⨁⨁◯◯ Low	Critical

Abbreviations: AE, adverse event; CI, confidence interval; HR, hazard ratio; MD, mean difference; RR, risk ratio; SAE, serious adverse event.

[a]O'Halloran et al [7] included patients 18 years or older. Fisher et al [11] included patients 16 years or older.

[b]Sample does not meet optimal information size, which suggests fragility with the estimate.

[c]95% CI cannot exclude the potential for no meaningful difference.

[d]Equivalent to World Health Organization categories 6, 7, or 8.

[e]Recovery rate ratio is equivalent to a hazard ratio.

[f]Some concerns with lack of allocation concealment and blinding, possibly leading to uneven administration of cointerventions remdesivir and corticosteroids.

[g]95% CI cannot exclude the potential for no meaningful difference or increased hospitalization with infliximab.

Regarding the definition of severe, rapidly progressing COVID-19, the panel recognizes that SpO_2 alone may not always correlate with disease severity, and the recommendation should be interpreted in the clinical context of the patient.

There are multiple biosimilars for infliximab, likely impacting the cost of the medication. In pregnant people, there are no RCTs on infliximab use in pregnancy; however, in available observational data on exposures in pregnancy, infliximab has not been associated with increased risk of miscarriage or birth defects [12]. Use in the third trimester has sometimes been minimized due to the concern of increased risk of infant infection in the first year of life, though registry data have not suggested this increased risk. Use of infliximab in the setting of COVID-19 is reasonable for pregnant patients if the benefits outweigh potential risks. Additionally, due to the large size of the molecule, minimal infant exposure via breastmilk would be expected [13–16].

CONCLUSIONS AND RESEARCH NEEDS

Head-to-head comparisons of baricitinib, tocilizumab, abatacept, and infliximab in patients with severe, rapidly progressing COVID-19 and critical COVID-19 would be informative. It is also uncertain whether a combination of 2 or more immunomodulatory agents (ie, baricitinib, tocilizumab, abatacept, infliximab) offers additional mortality or recovery benefits. In theory, infliximab (a TNF-α inhibitor) targets a different inflammatory pathway than tocilizumab (an IL-6 receptor blocker) or baricitinib (a JAK-STAT pathway inhibitor) and hence, may have a complementary benefit. Additionally, the efficacy of infliximab has not been evaluated in children or adolescents with severe or critical COVID-19. Future studies evaluating infliximab for COVID-19 should consider enrolling children and adolescents, particularly given the extensive pediatric experience with infliximab and available pediatric dosing recommendations for non-COVID-19 indications.

The guideline panel suggests the use of infliximab in hospitalized adults receiving systemic glucocorticoids who are experiencing severe, rapidly progressing COVID-19 or critical COVID-19, when baricitinib and tocilizumab are not available.

Supplementary Data

Supplementary materials are available at *Clinical Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

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Drs Adarsh Bhimraj and Rajesh T. Gandhi are chair and vice chair of the panel, respectively. The Hospitalized Patients subgroup, under the leadership of Dr Nandita Nadig, led the development of the recommendation. Remaining panelists assisted with interpretation of data, as well as drafting, revising, and approving the recommendation and manuscript. Drs Rebecca L. Morgan, lead methodologist, and Yngve Falck-Ytter, methodologist, were responsible for designing and performing the data analyses and leading the panel according to the GRADE process. Jennifer Loveless, methodologist, was responsible for project planning and management, including revisions to and final approval of the recommendation and manuscript.

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Potential conflicts of interest. Evaluation of relationships as potential conflicts of interest (COIs) is determined by a review process. The assessment of disclosed relationships for possible COIs is based on the relative weight of the financial relationship (ie, monetary amount) and the relevance of the relationship (ie, the degree to which an association might reasonably be interpreted by an independent observer as related to the topic or recommendation of consideration). No panelists had COIs directly related to infliximab. The following panelists have scientific advisory/consultant roles not related to the topic of COVID-19 with indicated companies: R. B. with Merck and Gilead, E. S. D. with Gilead, D. V. G. with Merck and Gilead, S. S. with Pfizer (concluded), P. T. with Merck, and R. T. G. with Merck. The following panelists have scientific advisory/consultant

roles related to COVID-19 but not severe/critical COVID-19 with indicated companies: R. B. with Shionogi, K. C. with Pardes Biosciences (concluded), A. Y. K. with Shionogi, S. S. with Adamis and Immunome (concluded), and P. T. with Shionogi. J. C. P. had a scientific advisory/consultant role (concluded) with InflaRx related to vilobelimab for critical COVID-19. No disclosures were reported for all other authors (the majority of panelists), including chair and vice chair.

All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

Additional information. More detailed information on the analysis and development of the recommendation is available in the Supplementary Material.

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